PROBLEM SETUP

Original LP:

$$
\begin{aligned}
\min \quad & -x_1 - 2x_2 \\
\text{s.t.} \quad & x_1 - x_2 \leq 1 \\
& 2x_1 - x_2 \geq -1 \\
& 2x_1 + x_2 \leq 5 \\
& x_1, x_2 \geq 0
\end{aligned}
$$

Standard Form: Introduce slack variables x_3, x_4, x_5:

$$
\begin{aligned}
\min \quad & -x_1 - 2x_2 \\
\text{s.t.} \quad & x_1 - x_2 + x_3 = 1 \\
& -2x_1 + x_2 + x_4 = 1 \\
& 2x_1 + x_2 + x_5 = 5 \\
& x_1, x_2, x_3, x_4, x_5 \geq 0
\end{aligned}
$$

Matrix Form:

$$
\mathbf{c} = \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}
$$

$$
\mathbf{A} = \begin{bmatrix} 1 & -1 & 1 & 0 & 0 \\ -2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 & 1 \end{bmatrix}
$$

$$
\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}
$$

General Framework: For standard form LP with basis $\mathbf{B}$, nonbasic $\mathbf{N}$:

Basic equations:

$$
\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} - \mathbf{B}^{-1}\mathbf{N}\mathbf{x}_N \tag{1}
$$

Objective:

$$
Z = \mathbf{c}_B^\top \mathbf{B}^{-1}\mathbf{b} + \bar{\mathbf{c}}_N^\top \mathbf{x}_N \tag{2}
$$

Reduced costs:

$$
\bar{\mathbf{c}}_N = \mathbf{c}_N^\top - \mathbf{c}_B^\top \mathbf{B}^{-1}\mathbf{N} \tag{3}
$$

Direction:

$$
\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{A}_j \tag{4}
$$

ITERATION EXAMPLE k=1

Initial Setup:

Basis $\mathbf{B}^1 = [\mathbf{A}_3, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Basis inverse $\mathbf{B}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Basic variables $\mathbf{x}_B = [x_3, x_4, x_5] = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$

Nonbasic variables $\mathbf{x}_N = [x_1, x_2] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Cost vectors: $\mathbf{c}_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, $\mathbf{c}_N = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$

Nonbasic matrix $\mathbf{N} = \begin{bmatrix} 1 & -1 \\ -2 & 1 \\ 2 & 1 \end{bmatrix}$

Reduced Cost Calculation: Using equation (5):

$$
\begin{aligned}
\bar{c}_1 &= c_1 - \mathbf{c}_B^\top \mathbf{B}^{-1}\mathbf{A}_1 \\
&= -1 - \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = -1 \\
\bar{c}_2 &= c_2 - \mathbf{c}_B^\top \mathbf{B}^{-1}\mathbf{A}_2 \\
&= -2 - \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = -2
\end{aligned}
$$

Objective Value: Using equation (4):

$$
\begin{aligned}
Z &= \mathbf{c}_B^\top \mathbf{B}^{-1}\mathbf{b} + \bar{\mathbf{c}}_N^\top \mathbf{x}_N \\
&= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} + \begin{bmatrix} -1 & -2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0
\end{aligned}
$$

Optimality Test: Current solution NOT optimal since $\bar{c}_1 = -1 < 0$ and $\bar{c}_2 = -2 < 0$.

By Bland's rule: choose smallest subscript among negative reduced costs. Choose x_1 to enter basis.

Direction Calculation: Using equation (6):

$$
\begin{aligned}
\mathbf{d}_B &= -\mathbf{B}^{-1}\mathbf{A}_1 \\
&= -\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix}
\end{aligned}
$$

Since $d_{B(2)} = 2 > 0$ and $d_{B(3)} = -2 < 0$, not unbounded.

Min-Ratio Test:

$$
\begin{aligned}
\theta^* &= \min_{i: d_{B(i)} < 0} \left\{ \frac{x_{B(i)}}{-d_{B(i)}} \right\} \\
&= \min \left\{ \frac{x_3}{-d_{B(1)}}, \frac{x_5}{-d_{B(3)}} \right\} \\
&= \min \left\{ \frac{1}{1}, \frac{5}{2} \right\} \\
&= \min\{1, 2.5\} = 1
\end{aligned}
$$

The basic variable x_3 exits the basis (smallest ratio).